Supplementary Appendix

Supplement to: Moonla C., Sosothikul D., Chiasakul T., et al. Anticoagulation and in-hospital mortality from coronavirus disease 2019: A systematic review and meta-analysis. 2021.

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Supplementary Table S1 Main characteristics of 37 included studies reporting the use of anticoagulants and in-hospital mortality in patients with coronavirus disease 2019 (COVID-19).

First author of the study	Study location	Mean age (yrs., ±SD)	No. of participants (%)							Non-survivor (N, %)
			Total	Female	ICU/ critically ill	No AC	Any AC	Prophylactic AC	Intermediate-to-therapeutic AC	
Al-Samkari H[27]	USA	61.7 ± 10.7	2,809	996 (35.5)	2,809 (100)	0 (0)	2,809 (100)	2,425 (86.3)	384 (13.7)	1,066 (37.9)
Alonso-Fernández A[28]	Spain	63.9 ± 9.2	30	11 (36.7)	11 (36.7)	1 (3.3)	29 (96.7)	26 (86.7)	3 (10)	0 (0)
Artifoni M[29]	France	61.7 ± 17.2	71	42 (59.1)	13 (18.3)	1 (1.4)	70 (98.6)	70 (98.6)	0 (0)	1 (1.4)
Ayerbe L[30]	Spain	65.3 ± 15.5	2,019	782 (38.7)	NR	285 (14.1)	1,734 (85.9)	NR	NR	286 (14.2)
Beyls C[31]	France	61.5 ± 6.0	12	2 (16.7)	12 (100)	0 (0)	12 (100)	0 (0)	12 (100)	2 (16.7)
Chen F[32]	China	52.3 ± 20.2	660	363 (55.3)	104 (15.8)	447 (67.7)	213 (32.3)	NR	NR	82 (12.4)
Contou D[33]	France	61.6 ± 17.2	26	19 (20.7)	26 (100)	0 (0)	26 (100)	NR	NR	13 (50)
Cui S[34]	China	59.9 ± 14.1	81	44 (54.3)	81 (100)	81 (100)	0 (0)	0 (0)	0 (0)	8 (9.9)
Desborough MJR[35]	UK	58.0 ± 10.1	66	18 (27.2)	66 (100)	0 (0)	66 (100)	55 (83.3)	11 (16.7)	20 (30.3)
Ferguson J[36]	USA	63.2 ± 10.9	141	63 (44.7)	141 (100)	0 (0)	141 (100)	95 (67.4)	46 (32.6)	40 (28.4)
Giacomelli A[37]	Italy	61.0 ± 13.1	233	72 (30.9)	8 (3.4)	216 (92.7)	17 (7.3)	0 (0)	17 (7.3)	48 (20.6)
Hanif A[38]	USA	62.0 ± 8.3	921	347 (37.7)	323 (35.1)	25 (2.7)	896 (97.3)	672 (73)	224 (24.3)	325 (35.3)
Helms J[39]	France	62.3 ± 10.7	150	28 (18.7)	150 (100)	0 (0)	150 (100)	105 (70)	45 (30)	13 (8.7)
Hsu A[40]	USA	63.3 ± 14.3	468	211 (45)	134 (28.6)	27 (5.8)	441 (94.2)	377 (80.6)	64 (13.7)	84 (17.9)
Ibañez C[41]	Spain	63.0 ± 10.7	19	9 (47.4)	19 (100)	1 (5.3)	18 (94.7)	18 (94.7)	0 (0)	4 (21.1)
Ionescu F[42]	USA	64.5 ± 17.0	3,480	1,796 (51.5)	642 (18.4)	361 (10.4)	3,119 (89.6)	2,121 (60.9)	998 (28.7)	506 (14.5)
Khalil K[43]	UK	66.9 ± 2.7	220	90 (40.9)	34 (15.5)	19 (8.6)	201 (91.4)	174 (79.1)	27 (12.3)	59 (26.8)
Klok FA[16]	Netherlands	64.0 ± 12.0	184	45 (24.5)	184 (100)	0 (0)	184 (100)	167 (90.8)	17 (9.2)	41 (22.3)
Lemos ACB[44]	Brazil	55.8 ± 13.3	20	4 (20)	20 (100)	0 (0)	20 (100)	10 (50)	10 (50)	7 (35)
Llitjos JF[45]	France	64.7 ± 13.7	26	6 (23.1)	26 (100)	0 (0)	26 (100)	8 (30.8)	18 (69.2)	3 (11.5)
Maatman TK[46]	USA	61.0 ± 16.0	109	47 (43.1)	109 (100)	0 (0)	109 (100)	109 (100)	0 (0)	27 (24.8)

Abbreviations: AC, anticoagulation; ICU, intensive care unit; NR, not reported; SD, standard deviation.

Supplementary Table S1 Main characteristics of 37 included studies reporting the use of anticoagulants and in-hospital mortality in patients with coronavirus disease 2019 (COVID-19). (continued)

First author of the study	Study location	Mean age (yrs., ±SD)	No. of participants (%)							Non-survivor (N, %)
			Total	Female	ICU/ critically ill	No AC	Any AC	Prophylactic AC	Intermediate-to-therapeutic AC	
Melmed KR[47]	USA	69.3 ± 16.2	416	149 (35.8)	136 (32.7)	271 (65.1)	145 (34.9)	0 (0)	145 (34.9)	126 (30.3)
Middeldorp S[48]	Netherlands	61.0 ± 14.0	198	68 (34.3)	75 (37.9)	0 (0)	198 (100)	111 (56.1)	87 (43.9)	38 (19.2)
Nadkarni GN[15]	USA	65.0 ± 14.3	4,389	1,932 (44)	467 (10.6)	1,530 (34.9)	2,859 (65.1)	1,959 (44.6)	900 (20.5)	1,073 (24.4)
Paolisso P[49]	Italy	67.0 ± 14.3	450	167 (37.1)	70 (15.6)	0 (0)	450 (100)	361 (80.2)	89 (19.8)	79 (17.6)
Pavoni P[50]	Italy	64.3 ± 12.1	42	15 (35.7)	42 (100)	0 (0)	42 (100)	0 (0)	42 (100)	9 (21.4)
Pesavento R[51]	Italy	70.7 ± 13.7	324	143 (44.1)	0 (0)	0 (0)	324 (100)	240 (74.1)	84 (25.9)	41 (12.7)
Pierce-Williams RAM[52]	USA	33.2 ± 5.8	64	64 (100)	44 (68.8)	17 (26.6)	47 (73.4)	37 (53.1)	10 (15.6)	0 (0)
Ranucci M[53]	Italy	60.3 ± 6.0	16	1 (6.3)	16 (100)	0 (0)	16 (100)	0 (0)	16 (100)	7 (43.8)
Tang N[14]	China	65.1 ± 12.0	449	181 (40.3)	97 (21.6)	350 (78)	99 (22)	99 (22)	0 (0)	134 (29.8)
Thomas W[54]	UK	59.0 ± 13.0	63	19 (30.2)	63 (100)	0 (0)	63 (100)	NR	NR	10 (15.9)
Travi G[55]	Italy	59.0 ± 9.5	13	4 (30.8)	9 (69.2)	0 (0)	13 (100)	13 (100)	0 (0)	3 (23.1)
Vizcaychipi MP[56]	UK	67.3 ± 16.0	939	376 (40)	NR	0 (0)	939 (100)	371 (39.5)	568 (60.5)	275 (29.3)
Wright FL[57]	USA	51.7 ± 10.1	44	16 (36.4)	44 (100)	0 (0)	44 (100)	44 (100)	0 (0)	5 (11.4)
Yuriditsky E[58]	USA	64.0 ± 8.3	64	18 (28.1)	64 (100)	0 (0)	64 (100)	9 (14.1)	55 (85.9)	19 (29.7)
Zeng DX[59]	China	61.0 ± 12.5	274*	129 (47.1)	59 (21.5)	208* (75.9)	66* (24.1)	66* (24.1)	0 (0)	47* (17.2)
Zerwes S[60]	Germany	63.7 ± 14.3	20	6 (30)	20 (100)	0 (0)	20 (100)	6 (30)	14 (70)	5 (25)

Abbreviations: AC, anticoagulation; ICU, intensive care unit; NR, not reported; SD, standard deviation.

* In the study by Zeng DX and colleagues, the mortality outcome for overall population was not stratified by the AC utilization. The meta-analysis included only a subgroup of elevated Padua score (N=89) which reported the outcome adequately.

Supplementary Table S2 Quality assessment for 36 included observational studies according to the Newcastle-Ottawa Scale (NOS)* for non-randomized studies.

First author of the study	Cohort design	Scored NOS†			Overall NOS	Quality grading‡
		Selection	Comparability	Outcome		
Al-Samkari H[27]	Retrospective	●●●●	●	●●●	●●●●●●●●	High
Alonso-Fernández A[28]	Prospective	●●●			●●●	Low
Artifoni M[29]	Retrospective	●●●		●	●●●●	Low
Ayerbe L[30]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Beyls C[31]	Retrospective	●●		●	●●●	Low
Chen F[32]	Retrospective	●●●●	●	●●●	●●●●●●●●	High
Contou D[33]	Retrospective	●●●		●●●	●●●●●●	Low
Cui S[34]	Retrospective	●●		●●●	●●●●●	Low
Desborough MJR[35]	Retrospective	●●●		●●●	●●●●●●	Low
Ferguson J[36]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Giacomelli A[37]	Prospective	●●	●	●●●	●●●●●●	Moderate
Hanif A[38]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Helms J[39]	Prospective	●●●●		●●●	●●●●●●●	Low
Hsu A[40]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Ibañez C[41]	Prospective	●●●		●	●●●●	Low
Ionescu F[42]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Khalil K[43]	Prospective	●●●●	●	●●●	●●●●●●●●	High
Klok FA[16]	Prospective	●●●●		●●●	●●●●●●●	Low
Llitjos JF[45]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Maatman TK[46]	Prospective	●●●		●●●	●●●●●●	Low
Melmed KR[47]	Retrospective	●●●	●	●●●	●●●●●●●	High
Middeldorp S[48]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Nadkarni GN[15]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Paolisso P[49]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Pavoni P[50]	Retrospective	●●●		●●●	●●●●●●	Low
Pesavento R[51]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Pierce-Williams RAM[52]	Retrospective	●●●	●	●	●●●●●	Low
Ranucci M[53]	Prospective	●●●		●	●●●●	Low
Tang N[14]	Retrospective	●●●●	●●	●●●	●●●●●●●●●	High
Thomas W[54]	Retrospective	●●●		●	●●●●	Low
Travi G[55]	Retrospective	●●		●●●	●●●●●	Low

* The assessment form is available online: Wells GA, Shea B, O'Connell D, et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analyses. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp

† A study can be given a maximum of 4, 2, and 3 stars (●) within the Selection, Comparability, and Outcome domains, respectively. A maximum of total 9 stars can be given for a study with the highest quality.

‡ Quality of each study is graded according to the following thresholds of NOS:
High quality: 3-4 stars in the Selection AND 1-2 stars in the Comparability AND 2-3 stars in the Outcome domains
Moderate quality: 2 stars in the Selection AND 1-2 stars in the Comparability AND 2-3 stars in the Outcome domains
Low quality: Less than 5 stars in overall NOS OR 0-1 star in the Selection OR 0 star in the Comparability OR 0-1 star in the Outcome domains

Supplementary Table S2 Quality assessment for 36 included observational studies according to the Newcastle-Ottawa Scale (NOS)* for non-randomized studies. (continued)

First author of the study	Cohort design	Scored NOS†			Overall NOS	Quality grading‡
		Selection	Comparability	Outcome		
Vizcaychipi MP[56]	Retrospective	●●●	●	●●●	●●●●●●●	High
Wright FL[57]	Prospective	●●●		●●	●●●●●	Low
Yuriditsky E[58]	Retrospective	●●●		●	●●●●	Low
Zeng DX[59]	Retrospective	●●●	●	●●●	●●●●●●●	High
Zerwes S[60]	Retrospective	●●●●		●●	●●●●●●	Low

* The assessment form is available online: Wells GA, Shea B, O'Connell D, et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analyses. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp

† A study can be given a maximum of 4, 2, and 3 stars (●) within the Selection, Comparability, and Outcome domains, respectively. A maximum of total 9 stars can be given for a study with the highest quality.

‡ Quality of each study is graded according to the following thresholds of NOS:
High quality: 3-4 stars in the Selection AND 1-2 stars in the Comparability AND 2-3 stars in the Outcome domains
Moderate quality: 2 stars in the Selection AND 1-2 stars in the Comparability AND 2-3 stars in the Outcome domains
Low quality: Less than 5 stars in overall NOS OR 0-1 star in the Selection OR 0 star in the Comparability OR 0-1 star in the Outcome domains

Supplementary Table S3 Risk of bias for the included randomized controlled trial assessed by the modified Cochrane Collaboration tool.[23]

First author of the study	Bias domain	Assessing element	Individual risk of bias	Overall risk of bias
Lemos ACB[44]	Selection bias	Random sequence generation	Low	Unclear*
		Allocation concealment	Low	
	Reporting bias	Selective outcome reporting	Low	
	Performance bias	Blinding of participants and personnel	Unclear	
	Detection bias	Blinding of outcome assessment	Low	
	Attrition bias	Incomplete outcome data	Low	
	Other bias	Bias due to problems not covered elsewhere	Low	

* The unclear overall risk of bias is for a trial that has at least 1 individual domain with unclear risk of bias, but without any domain with high risk of bias.

Supplementary Table S4 Patient characteristics from 17 studies included in the meta-analyses.

Characteristics	No./Total of participants (%) or Estimated values (95% CI)				
	Overall population	No AC	Any AC	Prophylactic dose AC	Intermediate-to-therapeutic dose AC
No. of participants (N)	17,833	3,547	14,286	8,792	3,547
Mean age (years)	63.7 (62.3-65.1)	62.5 (57.5-67.5)	65.6 (63.7-67.3)	64.1 (62.3-65.8)	65.8 (63.2-68.4)
Female gender	7,618/17,833 (42.7)	1,154/2,499 (46.2)	5,815/13,903 (41.8)	3,581/8,268 (43.3)	1,172/2,962 (39.6)
ICU/critically ill setting	5,036/14,875 (33.9)	104/798 (13.0)	4,217/8,379 (50.3)	3,169/6,462 (49.0)	1,048/1,917 (54.7)
Comorbidities					
Hypertension	6,441/13,602 (47.4)	799/2,162 (37.0)	5,366/10,691 (50.2)	3,489/7,124 (49.0)	1,377/2,628 (52.4)
Diabetes mellitus	4,076/13,861 (29.4)	491/2,189 (22.4)	3,453/10,923 (31.6)	2,237/7,348 (30.4)	862/2,636 (32.7)
Ischemic heart disease	1,601/12,672 (12.6)	237/2,189 (10.8)	1,297/9,823 (13.2)	823/7,243 (11.4)	474/2,580 (18.4)
Atrial fibrillation or flutter	710/11,544 (6.2)	182/2,162 (8.4)	528/9,382 (5.6)	182/6,866 (2.7)	346/2,516 (13.8)
Congestive heart failure	827/10,678 (7.7)	144/1,891 (7.6)	683/8,787 (7.8)	446/6,505 (6.9)	237/2,282 (10.4)
Cerebrovascular disease	364/4,140 (8.8)	23/361 (6.4)	289/3,119 (9.3)	171/2,121 (8.1)	118/998 (11.8)
Chronic pulmonary disease	1,727/10,475 (16.5)	212/1,918 (11.1)	1,466/7,808 (18.8)	754/4,818 (15.6)	391/2,051 (19.1)
Chronic liver disease	78/4,713 (1.7)	22/1,530 (1.4)	56/3,183 (1.8)	47/2,199 (2.1)	9/984 (0.9)
Chronic kidney disease	762/8,979 (8.5)	164/1,891 (8.7)	562/6,428 (8.7)	369/4,441 (8.3)	193/1,987 (9.7)
Malignancy	792/11,478 (6.9)	121/1,891 (6.4)	671/9,587 (7.0)	468/7,114 (6.6)	203/2,473 (8.2)
Current- or ex-smoker	1,548/11,387 (13.6)	158/1,891 (8.6)	1,320/9,263 (14.3)	925/6,874 (13.5)	395/2,389 (16.5)
Obesity	1,379/7,767 (17.8)	42/361 (11.6)	907/6,252 (14.5)	677/4,786 (14.1)	230/1,466 (15.7)
Prior venous thromboembolism	221/3,830 (5.8)	15/361 (4.2)	206/3,469 (5.9)	85/2,369 (3.6)	121/1,100 (11.0)
Immunocompromised status	74/4,409 (1.7)	25/1,530 (1.6)	49/2,879 (1.7)	39/1,969 (2.0)	10/910 (1.1)
Concomitant antiplatelets	871/7,775 (11.2)	131/1,530 (8.6)	734/6,012 (12.2)	577/4,634 (12.5)	157/1,378 (11.4)

Abbreviations: AC, anticoagulation; CI, confidence interval; ICU, intensive care unit.

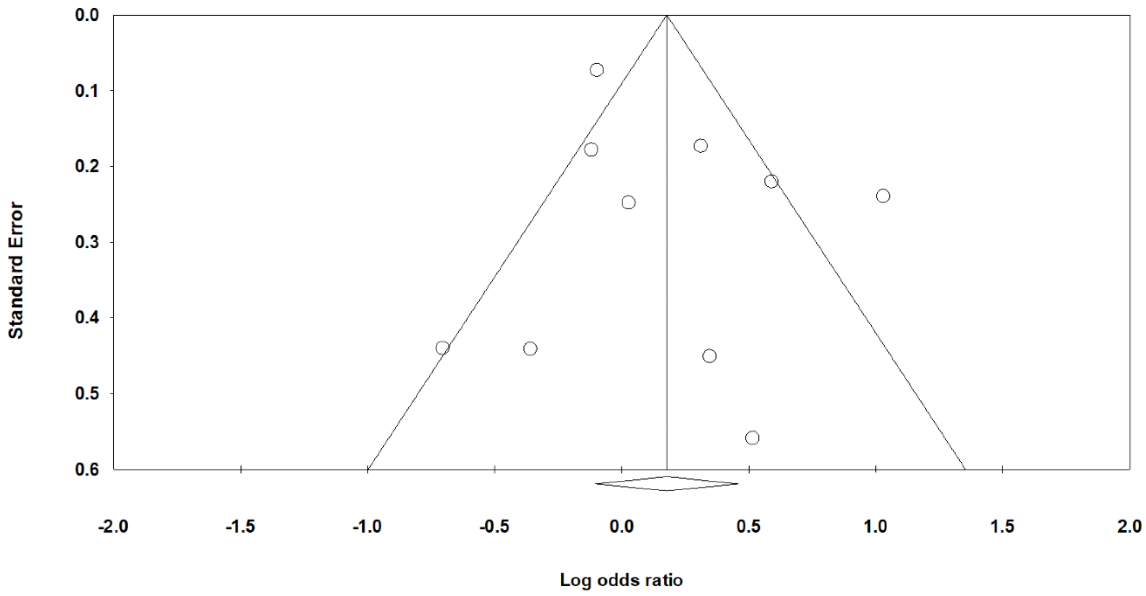
Supplementary Table S4 Patient characteristics from 17 studies included in the meta-analyses. (continued)

Characteristics	No./Total of participants (%) or Estimated values (95% CI)				
	Overall population	No AC	Any AC	Prophylactic dose AC	Intermediate-to-therapeutic dose AC
COVID-19 adjuvant treatments					
Remdesivir	23/485 (4.7)	N/A	23/485 (4.7)	14/345 (4.1)	9/140 (6.4)
Interleukin-6 receptor inhibitors	538/2,954 (18.2)	N/A	117/935 (12.5)	77/706 (10.9)	40/229 (17.5)
Corticosteroids	3,087/6,503 (47.5)	64/361 (17.7)	1,879/3,463 (54.3)	1,087/2,371 (45.8)	792/1,092 (72.5)
Hydroxychloroquine	5,132/6,434 (79.8)	127/361 (35.2)	3,001/4,054 (74.0)	2,096/2,827 (74.1)	1,052/1,227 (85.7)
Mean coagulation parameters and inflammatory markers					
D-dimer (μg/mL)	2.18 (1.68-2.68)	1.79 (1.20-2.38)	2.04 (1.35-2.73)	1.59 (0.88-2.29)	2.20 (1.39-3.00)
PT (seconds)	14.5 (13.8-15.2)	13.6 (13.5-13.8)	13.9 (13.1-14.7)	13.1 (10.9-15.4)	14.7 (14.1-15.3)
INR	1.17 (1.10-1.23)	1.30 (0.97-1.63)	1.19 (1.07-1.31)	1.09 (1.07-1.12)	1.22 (1.12-1.31)
aPTT (seconds)	34.9 (27.5-42.3)	28.1 (20.0-44.1)	35.8 (27.6-44.0)	29.1 (22.9-44.5)	37.7 (20.3-55.6)
Fibrinogen (g/L)	6.04 (4.34-7.74)	N/A	6.77 (6.50-7.03)	6.97 (6.53-7.41)	6.91 (6.27-7.55)
Platelets (x 10^9 cells/μL)	206.4 (191.9-220.8)	214.0 (210.5-217.4)	218.7 (204.4-232.9)	212.5 (198.7-226.3)	219.2 (200.7-237.8)
C-reactive protein (mg/L)	86.9 (53.4-120.4)	97.3 (93.3-101.3)	108.3 (61.4-155.2)	105.1 (56.5-153.7)	108.2 (39.0-177.5)
Procalcitonin (ng/mL)	0.20 (0.02-0.38)	0.27 (0.25-0.28)	0.28 (0.19-0.36)	0.25 (0.15-0.35)	0.27 (0.14-0.40)
Crude in-hospital mortality rates	4,211/17,833 (23.6)	763/3,547 (21.5)	3,448/14,286 (24.1)	2,037/8,792 (23.2)	1,125/3,547 (31.7)
Crude major bleeding rates	361/10,609 (3.4)	60/2,214 (2.7)	301/8,395 (3.6)	118/5,835 (2.0)	183/2,560 (7.1)

Abbreviations: AC, anticoagulation; aPTT, activated partial thromboplastin time; CI, confidence interval; COVID-19, coronavirus disease 2019; ESR, erythrocyte sedimentation rate; INR, international normalized ratio; N/A, not available; PT, prothrombin time.

Supplementary Figure S1 Funnel plots derived from standard errors and log odds ratios to determine risks of publication bias among the studies included in the comparative meta-analyses: (A) Any anticoagulation (AC) versus no AC, and (B) Intermediate-to-therapeutic dose versus prophylactic dose AC on mortality outcome. Asymmetry of funnel plot is statistically estimated by Egger's regression test. Abbreviation: CI, confidence interval.

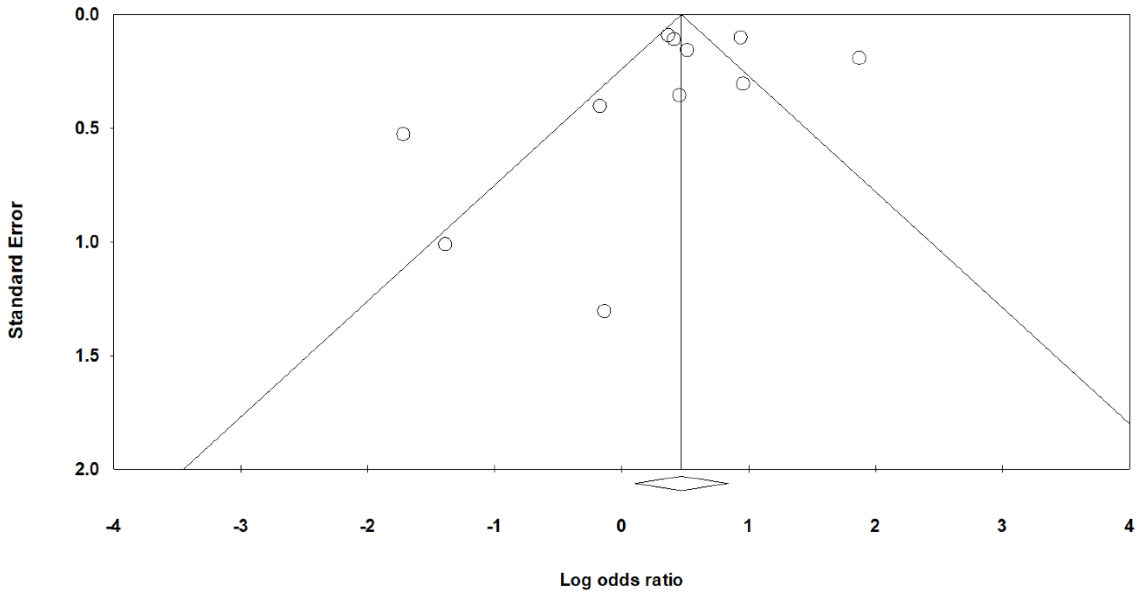
A. Anticoagulation *versus* no anticoagulation on mortality



No. of studies aggregated in the meta-analysis: 10 studies

Egger's regression intercept (random-effects model): 1.11 (95% CI -1.35 to 3.57), P = 0.328

B. Intermediate-to-therapeutic dose *versus* prophylactic dose anticoagulation on mortality

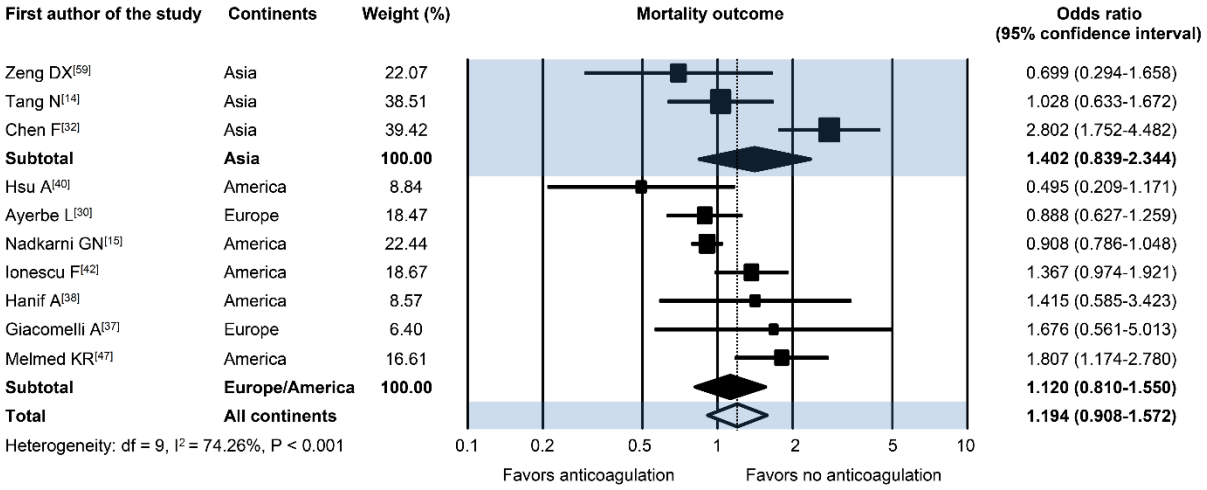


No. of studies aggregated in the meta-analysis: 11 studies

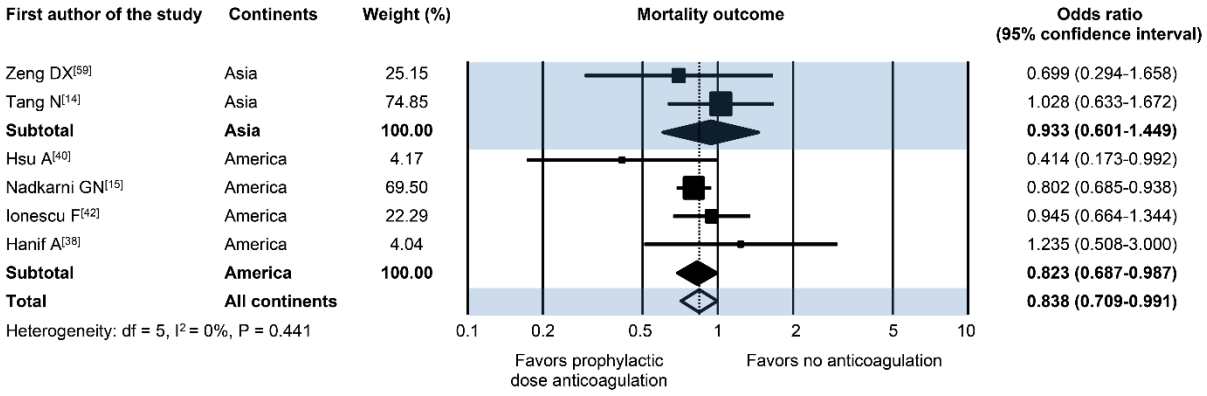
Egger's regression intercept (random-effects model): -1.06 (95% CI -4.73 to 2.60), P = 0.529

Supplementary Figure S2 The Forest plots present the pooled risk estimations for the effect of the anticoagulation (AC): (A) Any AC, and (B) Prophylactic dose AC, on in-hospital COVID-19 mortality, compared to no AC and *post hoc* stratified by study location subgroups (Asian versus European/American studies). Abbreviations: df, degree of freedom.

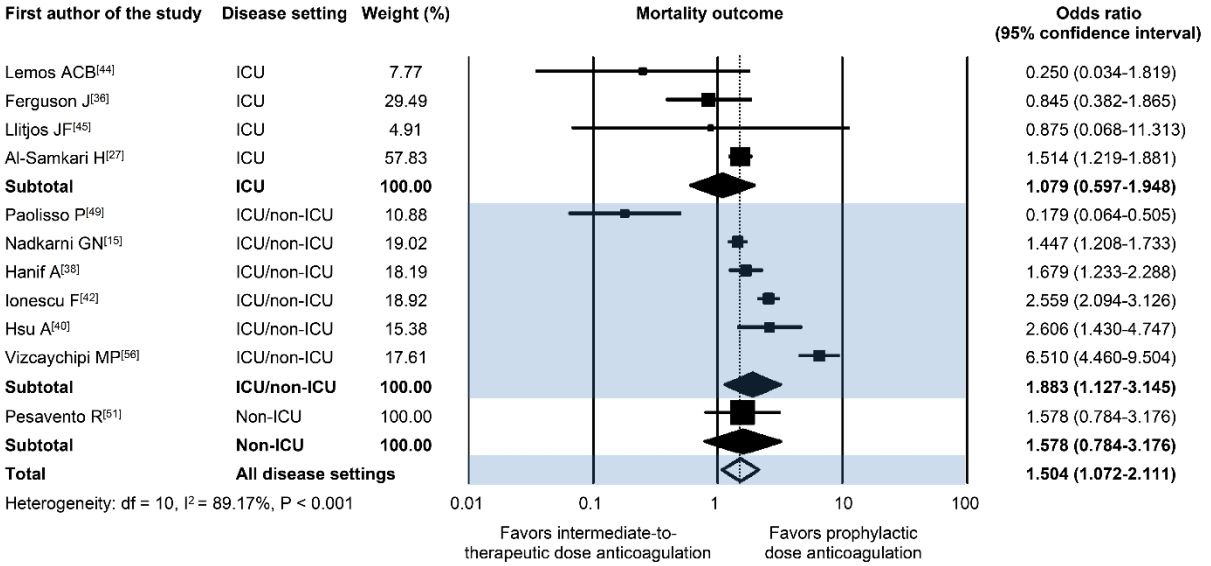
A. Anticoagulation *versus* no anticoagulation



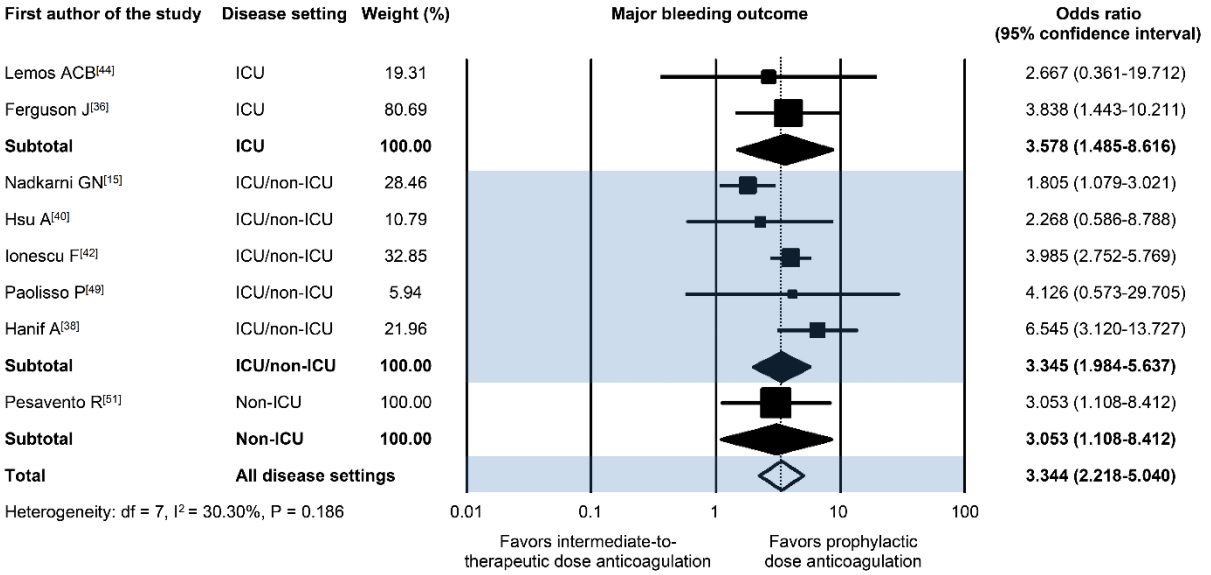
B. Prophylactic dose anticoagulation *versus* no anticoagulation



Supplementary Figure S3 The Forest plot presents the pooled risk estimations for the effect of the anticoagulation (AC) intensities (intermediate-to-therapeutic dose versus prophylactic dose AC) on in-hospital COVID-19 mortality, *post hoc* stratified by disease severity subgroups. Abbreviations: df, degree of freedom; ICU, intensive care unit.

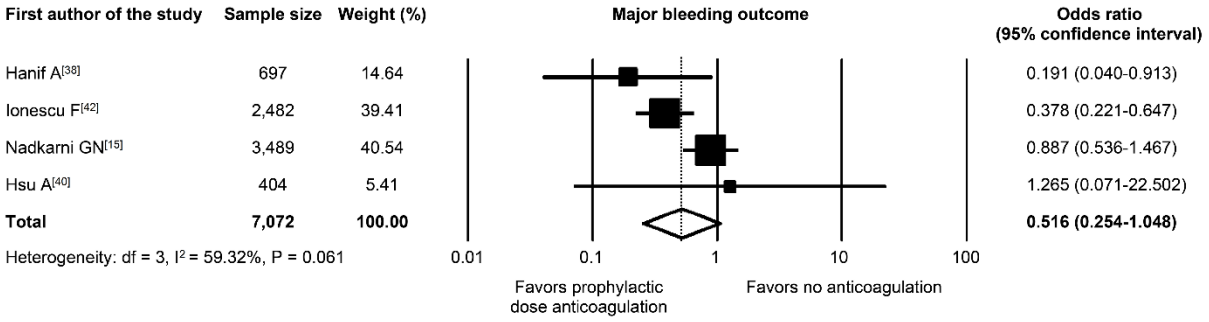


Supplementary Figure S4 The Forest plot presents the pooled risk estimations for the effect of the anticoagulation (AC) intensities (intermediate-to-therapeutic dose versus prophylactic dose AC) on major bleeding adverse events, *post hoc* stratified by disease severity subgroups. Abbreviations: df, degree of freedom; ICU, intensive care unit.



Supplementary Figure S5 The Forest plots present the pooled risk estimations for the effect of the anticoagulation (AC) intensities: (A) Prophylactic dose AC, and (B) Intermediate-to-therapeutic dose AC, on major bleeding adverse events, compared to no AC. Abbreviation: df, degree of freedom.

A. Prophylactic dose anticoagulation *versus* no anticoagulation



B. Intermediate-to-therapeutic dose anticoagulation *versus* no anticoagulation

